



Strathmore
UNIVERSITY

FACULTY OF INFORMATION TECHNOLOGY

BBT 2104 – INTEGRAL CALCULUS

Date: Tuesday 5th July 2016

Time: 2 Hours

Instruction

Answer QUESTION ONE and any other TWO questions

QUESTION ONE (30 Marks)

- a) Evaluate the given integral $\int (3x^2 + 1)(x^3 + x)^5 dx$ (3 marks)
- b) Determine the area between the curve $y = x(4 - x)$ and the $x - axis$ from $x = 0$ to $x = 5$. (4 marks)
- c) Use an appropriate method to evaluate
 - i. $\int \sin^3 x \cos^2 x dx$ (5 marks)
 - ii. $\int_0^{\frac{\pi}{4}} (1 + \cos 4x)^{0.5} dx$ (4 marks)
- d) Find the general solution of the differential equation $x \frac{dy}{dx} + 2 \frac{dy}{dx} - 1 = y$ (4 marks)
- e) Determine the volume of revolution formed by rotating the area enclosed by the curve $y = x + x^2$, the $x - axis$ and the points $x = 2$ and $x = 3$ through one revolution about the $x - axis$. (3 marks)
- f) Use partial fractions to evaluate $\int \frac{x^2 + 4x + 1}{(x-1)(x+1)(x+3)} dx$ (4 marks)
- g) Use Taylor's theorem to obtain a series expansion for $\cos(x + \frac{\pi}{3})$ up to and including terms in x^3 . (3 marks)

QUESTION TWO (20 Marks)

- a) Find the area between the line $y = x - 1$ and the curve $y^2 = 2x + 6$ (7 marks)
- b) Given the function $f(x) = \sin x$ over the range $x = 0$ to $x = 2\pi$. Compute;
 - i. The definite integral of the function over the given range. (3 marks)

- ii. The area between the graph of $f(x)$ and the $x - axis$ over the same range. (4 marks)
- c) Find the area of the region bounded by the curve $ay^2 = x^3$, the $y - axis$ and the lines $y = a$ and $y = 2a$ (6 marks)

QUESTION THREE (20 Marks)

- a) Use an appropriate method to evaluate the following integrals;
- i. $\int \cos^5 x \, dx$ (5 marks)
 - ii. $\int \cos^4 x \sin^2 x \, dx$ (6 marks)
 - iii. $\int e^x \cos x \, dx$ (6 marks)
 - iv. $\int \ln x \, dx$ (3 marks)

QUESTION FOUR (20 Marks)

- a) Derive the reduction formula for $\int \sin^n x \, dx$ hence evaluate $\int \sin^3 x \, dx$ (10 marks)
- b) Using the identity $\tan^2 x + 1 = \sec^2 x$ evaluate $\int \tan^4 x \, dx$ (5 marks)
- c) Use an appropriate trigonometric substitution to evaluate $\int \frac{x^2 \, dx}{(9-x^2)^{0.5}}$ (5 marks)

QUESTION FIVE (20 Marks)

- a) Determine the volume of the solid of revolution formed by revolving the area enclosed by the curve $xy = 3$, the $x - axis$ and the points $x = 2$ and $x = 3$ through one revolution about the $x - axis$. (5 marks)
- b) Determine the volume of the solid of revolution formed by revolving the area enclosed by the curve $x^2 + y^2 = 16$, the $y - axis$ and the points $y = 0$ and $y = 4$ through one revolution about the $y - axis$. (5 marks)
- c) Solve the following differential equations;
- i. $(x + 2) \frac{dy}{dx} = (2x^2 + 4x + 1)(y - 3)$ given that $x = 0$ when $y = 7$ (7 marks)
 - ii. $\frac{dy}{dx} = \frac{2}{\sin y}$ (3 marks)